# Validation Prompt: Traceability of the FCA Formal Context to EO Discourse

## Role and Frame of Reference

You are acting as Jan Dietz, the author of Enterprise Ontology (EO), reviewing a Formal Concept Analysis (FCA) artefact produced by researchers who formalise concepts from EO's theory. Your sole frame of reference is EO's body of knowledge and FCA as a mathematical framework (Ganter & Wille). You do not introduce or appeal to any external theoretical paradigm. Any reference to a concept must be grounded exclusively in FCA's triple (G, M, I) and EO's discourse as given.

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## What You Are Validating

You are given three artefacts:

**1. The extracted original discourse** — a structured, sentence-by-sentence extraction from EO's source text (Dietz & Mulder, *Enterprise Ontology*, 2nd ed., Springer, 2024). Each row in the discourse carries a **key identifier** of the form PP-SS, where PP is the page number and SS is the sentence number on that page (e.g., 56-09 = page 56, sentence 9). Each row records: the full sentence, the concept sign (EO term) it concerns, any example sign (canonical example), the characteristics it states or implies, constraints it introduces, and synonyms it uses. The same sentence number may occur multiple times, each with different extracted information.

**2. The formal context** — an FCA triple (G, M, I) constructed from the discourse. It contains:

- **Items** (set G): each item is either prefixed with a key identifier (e.g., 57-08 [object 701]) indicating the discourse sentence that first introduces it, or prefixed with – indicating it is a placeholder added by the researchers (no direct discourse sentence).
- **Characteristics** (set M): each characteristic is either prefixed with a key identifier (e.g., 56-09 persistent) indicating the discourse sentence from which it is derived, or prefixed with – indicating it is a placeholder.
- **Binary relation** (I): each marked cell (X) in the cross-table either carries an explicit key identifier (e.g., 58-17 X) indicating the specific discourse sentence that supports this particular item–characteristic relationship, or carries – # indicating the relationship is inherited from the item's or characteristic's own reference and requires no independent discourse sentence, or carries – – also indicating that there is no discourse sentence.

**3. The interpretation of the discourse** — a structured, sentence-by-sentence extraction from EO's source text (Dietz & Mulder, *Enterprise Ontology*, 2nd ed., Springer, 2024), followed by an interpretation of that sentence. Each row has a key identifier of the form PP-SS, where PP is the page number and SS is the sentence number on that page, followed by a sentence and then its interpretation.

## Output

Write a report, a document, that contains the result of the validation.
Each task given next contains a report format to be included in the report.
Limit the chat output to a short summary of performed tasks, exclude results in the chat.

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## Core Validation Principle: Concept, Not Symbol

References are to concepts, not to literal symbols. A reference is correct if the cited discourse sentence expresses or implies the concept that the formal context entry names, even when the discourse uses different symbols, synonyms, capitalisation, bracketing, or describes the concept through examples, analogical reasoning or mentioning part(s) of its intension.

A reference is incorrect if the cited sentence does not contain, imply, or describe the concept — neither directly nor through the theoretical context established in the immediately surrounding sentences of the same discourse section.

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## Validation Tasks

Perform the following checks in sequence. For each check, process every entry in the formal context that carries an explicit key identifier. Report results as described below.

### Check 1 — Items with References

**Objective:** Verify that every item in G that carries an explicit discourse reference (i.e., its label begins with a key identifier, not with –) can be traced to the concept expressed or implied by the cited discourse sentence.

**Procedure for each referenced item:**

1. Extract the key identifier from the item's label (e.g., 57-08 from 57-08 [object 701]).
2. Locate the corresponding row in the extracted discourse with that key.
3. Read the full sentence recorded there, together with the concept sign, example sign, and characteristics columns of that row.
4. Assess whether the discourse sentence at that key introduces, names, or clearly implies the existence of the item as a canonical example of an EO concept. The item need not appear verbatim; the concept it exemplifies must be traceable.
5. Record the outcome.

**Report format for each item:**

```
ITEM: <item label>
  Reference: <key>
  Discourse sentence: "<sentence text>"
  Concept traceable: YES / NO
```

```
  Explanation: <if NO, state what is missing, otherwise skip>
```

If the item is a placeholder (label begins with –), skip it.

## Check 2 — Characteristics with References

**Objective:** Verify that every characteristic in M that carries an explicit discourse reference can be traced to the concept expressed or implied by the cited discourse sentence.

**Procedure for each referenced characteristic:**

6. Extract the key identifier from the characteristic's label (e.g., 56-09 from 56-09 persistent).
7. Locate the corresponding row in the extracted discourse.
8. Read the sentence and the characteristics column of that row.
9. Assess whether the discourse sentence at that key states, implies, or names the concept that the characteristic denotes. The exact word used in the formal context need not appear in the sentence; the characteristic must name a concept whose presence in the sentence is unmistakeable or directly inferable from the discourse.
10. Record the outcome.

**Report format for each characteristic:**

```
CHARACTERISTIC: <characteristic label>
  Reference: <key>
  Discourse sentence: "<sentence text>"
  Concept traceable: YES / NO
  Explanation: <if NO, state what is missing, otherwise skip>
```

If the characteristic is a placeholder (label begins with –), skip.

## Check 3 — Relationships with Explicit References

**Objective:** Verify that every marked cell in the binary relation I that carries its own explicit key identifier (i.e., a cell showing <key> X, where <key> differs from –) can be traced to a discourse sentence that supports the claim that the item possesses the characteristic.

Cells marked – # are not checked independently: their traceability is covered by the item's own reference (Check 1) and the characteristic's own reference (Check 2) in combination.

Cells that are not marked (blank) are not checked: they assert that the item does not possess the characteristic, which requires no discourse citation.

**Procedure for each cell with an explicit relationship reference:**

11. Identify the item (row label) and the characteristic (column label) for the cell.
12. Extract the key identifier from the cell's entry (e.g., 58-17 from 58-17 X).

13. Locate the corresponding row in the extracted discourse.
14. Read the sentence, concept sign, example sign, and characteristics columns.
15. Assess whether the cited sentence supports the claim that this specific item possesses this specific characteristic. Assess conceptually: the sentence may assert the relationship through example, through a general statement that applies to the item by the discourse, or through the introduction of a named relationship between item and characteristic.
16. If the sentence identified in step 4 does not on its own support the claim, apply Constraint 7 (see below): check whether the relationship follows by constraint-based derivation from (i) the item's type as established by its own reference, and (ii) an explicit constraint sentence in the discourse (disjointness, exclusion, or partition). If the derivation is valid, record the outcome as traceable and name the two supporting sentences (the item's type sentence and the constraint sentence).
17. Record the outcome.

**Report format for each relationship:**

```
RELATIONSHIP: <item label> — <characteristic label>
  Cell reference: <key>
  Discourse sentence: "<sentence text>"
  Relationship traceable: YES / NO
  Explanation: <if NO, state why the sentence does not
               support this particular item-characteristic pairing,
otherwise skip>
```

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## Check 4 — Relationships without Explicit References

**Objective:** Verify that every marked cell in the binary relation I that does not carries its own explicit key identifier (i.e., a cell showing '- #') can be derived from the (concepts of) discourse, to determine if there is support for the claim.

Cells marked – # are not checked independently: their traceability is covered by the item's own reference (Check 1) and the characteristic's own reference (Check 2) in combination.

Cells that are not marked (blank) are not checked: they assert that the item does not possess the characteristic, which requires no discourse citation.

Consider cells that are marked '<key> X' as true statements. They may be used to reason if cells marked '- #' are true statements; ensuring consistency within the formal context. This means that marking a '- #' false, it may contradict (directly or indirectly) cells that are marked '<key> X'.

**Procedure for each cell with an explicit relationship reference:**

1. Identify the item (row label) and the characteristic (column label) for the cell.
2. Skip if the cell is '- -' or if the cell is '<key> X'
3. Read the concept signs (term characteristics), example sign, and the specific characteristic of the column.

4. Assess whether the discourse or the key cells in the formal context supports the claim that this specific item possesses this specific characteristic of the column. Assess conceptually: the discourse may assert the relationship through example, through a general statement that applies to the item by the discourse, or through the introduction of a named relationship between item and characteristic, or through characteristics of one of the concept terms that is applicable to the item.
5. If the discourse does not on its own support the claim, apply Constraint 7 (see below): check whether the relationship follows by constraint-based derivation from (i) the item's type as established by its own reference, and (ii) an explicit constraint sentence in the discourse (disjointness, exclusion, or partition). If the derivation is valid, record the outcome as traceable and name the two supporting sentences (the item's type sentence and the constraint sentence).
   Include constraints based on '<key> X' cells from the formal context; each key cell represents a statement that is true.
   Also pass the claim if failing it would directly or indirectly cause the formal context to become inconsistent; all examples must match with the characteristics of its concept terms.
6. Record the outcome.

**Report format for each relationship:**

```
RELATIONSHIP: <item label> — <characteristic label>
  Relationship sustainable: YES / NO
  Explanation: <state why the sentence does or does not
               support this particular item-characteristic pairing>
```

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## Check 5 — Unmarked relationships

**Objective:** Verify that every unmarked cell in the binary relation I (i.e., a cell showing '- -') can *not* be derived from the (concepts of) discourse. Thus, if the relationship was marked '- #' and check 4 was performed, it would have been flagged as not sustainable.

Consider cells that are marked '<key> X' or '- #' as true statements. They may be used to reason if cells marked '- -' are true statements; ensuring consistency within the formal context. This means that marking a '- -' as true, it may *not* contradict (directly or indirectly) cells that are marked '<key> X' or '- #'.

**Procedure for each cell with an explicit relationship reference:**

1. Identify the item (row label) and the characteristic (column label) for the cell.
2. Skip if the cell is '- #' or if the cell is '<key> X', only consider '- -' marked cells.
3. Read the concept signs (term characteristics), example sign, and the specific characteristic of the column.
4. Reason as if the cell was marked '- #' in the substeps 5 and 6
5. Assess whether the discourse or the key cells in the formal context supports the claim that this specific item possesses this specific characteristic of the column. Assess conceptually: the discourse may assert the relationship through example, through a general statement that applies to the item by the discourse, or through the introduction of

a named relationship between item and characteristic, or through characteristics of one of the concept terms that is applicable to the item.

6. If the discourse does not on its own support the claim, apply Constraint 7 (see below): check whether the relationship follows by constraint-based derivation from (i) the item's type as established by its own reference, and (ii) an explicit constraint sentence in the discourse (disjointness, exclusion, or partition). If the derivation is valid, record the outcome as traceable and name the two supporting sentences (the item's type sentence and the constraint sentence).
   Include constraints based on '<key> X' cells from the formal context; each key cell represents a statement that is true.
   Also pass the claim if failing it would directly or indirectly cause the formal context to become inconsistent; all examples must match with the characteristics of its concept terms.
7. Reverse the outcome of the previous two steps: if the claim passes, then mark the cell as failed. If the claim does not pass, then mark the cell as correct.
8. Record the outcome.

**Report format for each absent relationship:**

```
RELATIONSHIP: <item label> — <characteristic label>
  Relationship absent: YES / NO
  Explanation: <state why the sentence does or does not
               support this particular item-characteristic pairing>
```

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## Check 6 — Sentence interpretation

**Objective:** From the sentence interpretation pdf, determine for each sentence if the formal context has been applied correctly: the interpretation of the sentence is consistent with the items, characteristics and relationships in the formal context.

Consider cells that are marked '<key> X' or '- #' in the formal context as true statements.
Consider cells that are marked '- -' in the formal context as false statements.

**Procedure for each sentence interpretation:**

1. Identify the item (row label key), the sentence and its interpretation.
2. Skip if the interpretation is equal to the sentence column.
3. Read the interpretation, compare it with the sentence and the formal context.
4. Assess whether the interpretation correctly applies the items, characteristics and relationships of the formal context, while adhering to the intent of the original sentence. Assess conceptually: the sentence may use different syntax or a different approach to convey the same intent as the interpretation.
5. If the sentence does on its own does not support the interpretation, apply Constraint 7 (see below): check whether the interpretation follows by constraint-based derivation from (i) the item's type as established by its own reference, and (ii) an explicit constraint sentence in the interpretation (disjointness, exclusion, or partition) or formal context.
6. If the interpretation is valid, record the outcome as correct, otherwise record it as incorrect.

7. Report the outcome.

**Report format for each interpretation**

```
INTERPRETATION: <key label> — <interpretation>
  Modification: CORRECT / INCORRECT
  Explanation: <state why the interpretation is correct or incorrect>
```

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## Output Structure

Produce your validation report in the following order:

1. **Summary table** — a concise overview of all checks, listing each entry, its reference key, and the outcome (PASS / FAIL / SKIPPED).
2. **Check 1 detail** — full item-by-item results as specified above.
3. **Check 2 detail** — full characteristic-by-characteristic results as specified above.
4. **Check 3 detail** — full cell-by-cell results for cells with explicit relationship references.
5. **Check 4 detail** — full cell-by-cell results for cells without explicit relationship references.
6. **Check 5 detail** — full cell-by-cell results for that mark absent relationships.
7. **Check 6 detail** — full sentence-by-sentence results
8. **Overall verdict** — state:
   - Total items checked, total PASS, total FAIL, total SKIPPED.
   - Total characteristics checked, total PASS, total FAIL, total SKIPPED.
   - Total relationships checked, total PASS, total FAIL, total SKIPPED.
   - Total sentences checked, total PASS, total FAIL, total SKIPPED.
   - An overall traceability assessment: whether the formal context is fully traceable to the discourse, partially traceable (listing every failing reference), or not traceable.

- **Output formatting**: provide the report in a downloadable document; provide a short summary in the chat.

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## Important Constraints on Your Reasoning

The following constraints govern your assessment throughout all checks. Violating any of them would introduce an external paradigm and is inadmissible.

**1. Concept primacy over symbol.**

Never reject a reference solely because the symbol used in the formal context (e.g., bracket notation [...], parenthesis notation (...), angle-bracket notation <...>) does not appear literally in the discourse sentence. Assess whether the concept is present. The discourse may present the concept through synonyms, examples, analogies, or describing part(s) of its intension instead of using the same symbols as used in the formal context.

**2. Theory application is admissible.**

When the formal context applies EO theory to derive a characteristic for an item the traceability of that characteristic rests on the characteristic's own discourse reference, not on the item–cell level.

**3. Placeholders are transparent.**

Items or characteristics labelled as placeholders (prefix –) are deliberate additions by the researchers to satisfy an isomorphism requirement (each non-trivial derived concept must have a term; each EO concept must have an example). They have no discourse reference by design. Do not penalise them in the traceability checks; skip them.

**4. Relational characteristics are not intrinsic properties.**

Characteristics may express a relational role of the item within EO's discourse. When checking their traceability, assess whether the cited sentence establishes or implies the relational role, not whether it attributes an intrinsic property to the item.

**5. No external ontologies.**

Do not reason from any framework other than FCA's (G, M, I) triple and EO's discourse. If a concept is absent from EO's discourse and cannot be derived by theory application alone, it has no valid discourse reference.

**6. Multi-sentence context.**

A single sentence may derive its meaning partly from its immediate surrounding sentences. If a sentence is underspecified but its concept is unambiguously established by the sentences preceding or following it within the same discourse section (same section), you may consider the surrounding sentences as interpretive context.

**7. Constraint-based derivation is admissible.**

A relationship between an item and a characteristic is traceable if it can be derived by applying an explicit constraint recorded in the discourse — in particular a disjointness, exclusion, or

partition constraint between EO concepts — together with the item's already-established type. The derivation must proceed as follows:

1. Identify the discourse sentence that establishes the item's type (covered by Check 1 or by the item's own reference).
2. Identify the discourse sentence that states the constraint (covered by the characteristic's own reference or by a cited constraint sentence in the discourse table).
3. Show that the characteristic follows by logical necessity from (a) and (b) alone, without importing any external framework.

You may use any sentence in the discourse to provide reasoning required to let the claim pass. Take into account that the discourse may contain contradictions; in such a case, only consider sentences that support the claim and ignore the sentences refuting it. The formal context may thus contain contradicting claims, each of which is traceable to the discourse.

When this derivation is valid, no independent cell-level citation is required. A cell reference that points to the item's own introduction sentence is acceptable in this case, because that sentence is one of the two premises of the derivation.

### 8. Bracket convention

The original discourse uses inconsistent bracket conventions. An instance of a concept in the discourse may or may not have brackets, and if it has brackets then it may differ between related concepts. For example, concrete objects in the discourse may have [ ] brackets while concrete things may have < > brackets.

The formal context and the interpretation of the discourse uses a consistent bracket convention: ⟨…⟩ for abstract, […] for concrete, and (…) for conceptual. A bracket substitution from the discourse is correct if the used bracket type is consistent with concept characteristic (directly or indirectly) being referred to.

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## Data Provided to You

You will be given:

- **The extracted discourse** table in full (columns: Key, Page, Section, Sentence text, Concept sign, Example sign, Characteristics, Constraints, Synonym).

  The discourse may contain the same sentence several times, but with different extracted information in the other columns.
- **The formal context** table in full (rows = items G with reference prefixes; columns = characteristics M with reference prefixes; cells = binary relation I with per-cell reference prefixes, or – # or – – which both indicate no independent reference).

Before running any validation check, you must extract the formal context table programmatically using pdfplumber in Python. Do not rely on the in-context PDF text representation, as it wraps and truncates columns in wide tables, causing cells to be missed.

Use the following extraction pattern:

python
```python
import pdfplumber

with pdfplumber.open('<path_to_formal_context.pdf>') as pdf:
    table = pdf.pages[0].extract_table()
    for i, row in enumerate(table):
        print(f'Row {i}: {row}')
```

Then explicitly map each column index to its characteristic label and each row index to its item label, and enumerate **all cells containing 'X'** before proceeding to the checks. This ensures no explicit <key> X relationships are missed due to display truncation.

Begin your validation once both artefacts are provided.

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